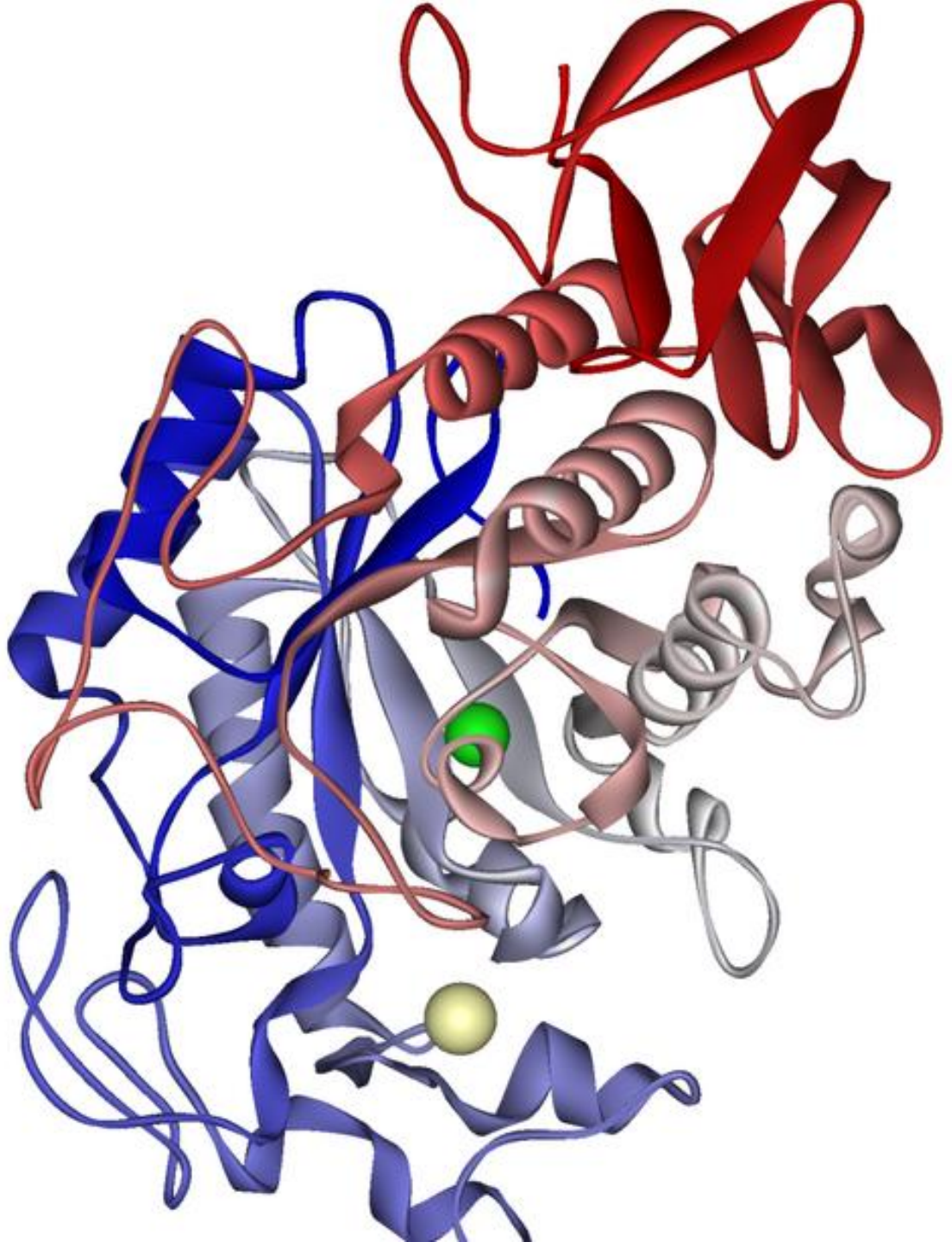
DP Biology – C1.1 (HL extension)

Controlling metabolism

- How do enzymes work with other molecules?
- Which parts of metabolism depend on one another?



HL statements C1.1.11–17. Textbook pp. 396–404.
Image: salivary amylase, an extracellular enzyme. Public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.1.11–13	Compare intracellular and extracellular reactions, explain why metabolism releases heat, and compare linear and cyclical pathways
2	C1.1.14–15	Compare non-competitive (allosteric) and competitive inhibition, using statins, and interpret data
3	C1.1.16	Explain feedback inhibition, using the pathway that makes isoleucine
4	C1.1.17	Explain mechanism-based inhibition by penicillin, and how resistance arises

Where enzymes work

C1.1.11

	Intracellular	Extracellular
Where	Inside the cell that made them: cytoplasm, organelles, membranes	Outside the cell that made them
Examples	Glycolysis (cytoplasm); Krebs cycle (mitochondrial matrix)	Digestion in the gut by amylase, lipase and proteases
Made by	Mostly free ribosomes	Ribosomes on rough ER; packaged in the Golgi and secreted by exocytosis

Think about it

Fungi digest their food outside their bodies, and so do many bacteria. Suggest why secreting enzymes is useful for an organism that lives in its food.

Metabolism always releases heat

C1.1.12



Emperor penguins. Ian Duffy, CC BY 2.0.

No energy transfer in metabolism is 100% efficient: some energy is always lost as **heat**. Aerobic respiration, for example, stores only about a third of the energy in glucose as ATP.

Endotherms (mammals and birds) use this heat to keep a constant body temperature. Shivering raises heat output; **brown fat** releases heat directly from respiration.



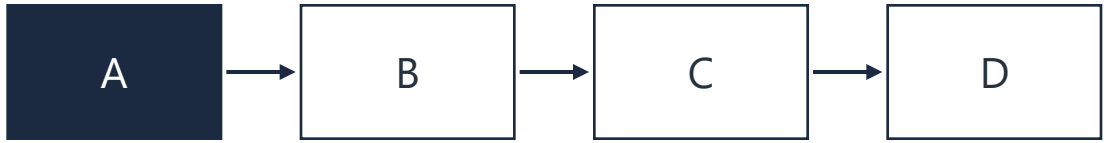
A steaming compost heap. Albarubescens, CC BY-SA 4.0.

Question

Compost heaps can reach 60 °C. Which organisms release this heat, and why does a large heap get hotter than a small one?

Linear and cyclical pathways

C1.1.13



Linear: glycolysis (glucose to pyruvate)

Key idea

Each step has its own enzyme, so each step can be controlled.

	Linear	Cyclical
Shape	A chain from a starting substrate to an end product	The last step regenerates the first substrate
Examples	Glycolysis	Krebs cycle; Calvin cycle
Feature	Each intermediate is used once	A small amount of carrier (oxaloacetate, RuBP) handles many molecules

Non-competitive inhibition at an allosteric site

C1.1.14

An **allosteric site** is a binding site on an enzyme **away from the active site**. Only specific molecules fit it.

When an inhibitor binds there, the shape of the enzyme changes, so the active site no longer catalyses the reaction efficiently.

Binding is usually **reversible**. Because the inhibitor does not compete for the active site, adding more substrate **does not** overcome the inhibition.

Why "non-competitive"?

The inhibitor does not compete with the substrate for the active site. Both can be bound at once, but the enzyme cannot convert the substrate. So the maximum rate is lowered.

Competitive inhibition: statins

A **competitive inhibitor** has a shape similar to the substrate and binds reversibly to the **active site**, blocking it.

Increasing the substrate concentration makes it more likely that substrate, not inhibitor, binds, so the maximum rate can still be reached.

Statins resemble HMG-CoA and inhibit **HMG-CoA reductase**, an enzyme in cholesterol synthesis in the liver.

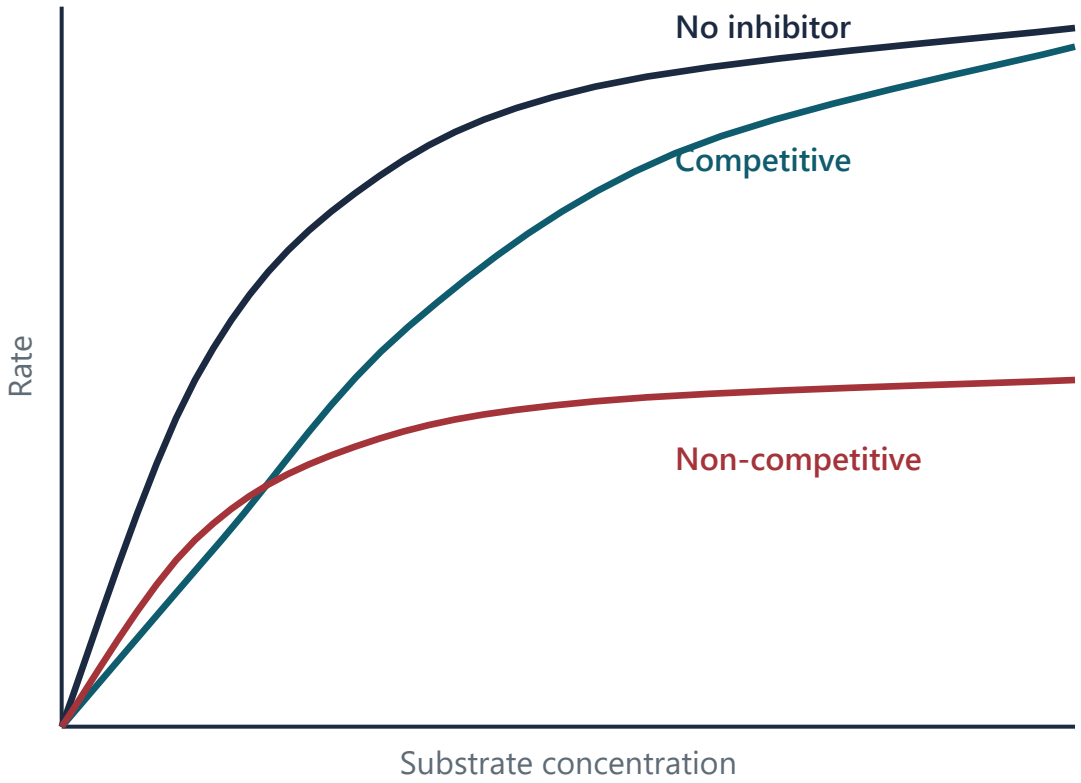
Statins in medicine

Less cholesterol is made, so LDL cholesterol in the blood falls. This lowers the risk of atherosclerosis and coronary heart disease.

	Competitive	Non-competitive
Binds to	Active site	Allosteric site
Effect of more substrate	Overcomes inhibition	No effect on inhibition

Telling inhibitors apart from data

C1.1.14–15



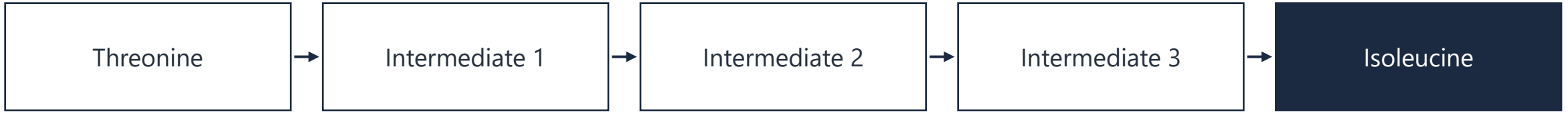
Substrate	None	X	Y
2	40	15	20
10	88	62	44
40	100	97	50

Activity

Invented data: rates as % of the maximum without inhibitor. Which inhibitor, X or Y, is competitive? Justify your answer from the data at high substrate concentration.

Feedback inhibition: isoleucine

C1.1.16



Enzyme 1, threonine deaminase, is inhibited when isoleucine binds to its allosteric site.

When isoleucine builds up, it binds threonine deaminase and slows the whole pathway.

When isoleucine is used up (for protein synthesis), it leaves the allosteric site and the pathway restarts.

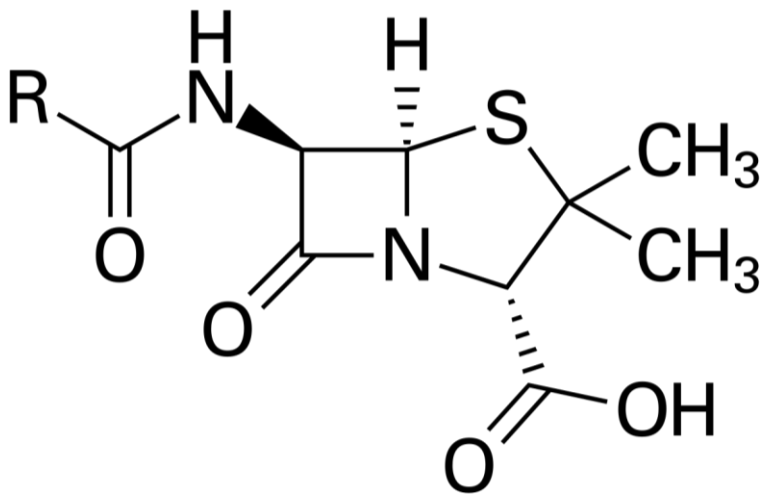
This is **negative feedback**: the end product controls its own production, so the cell does not waste energy or threonine making isoleucine it does not need.

Why the first enzyme?

Inhibiting the first step stops all the intermediates building up. Inhibiting a later step would waste the earlier steps.

Mechanism-based inhibition: penicillin

C1.1.17



Core structure of penicillin. Public domain.

Bacteria use **transpeptidases** to cross-link the peptidoglycan in their cell walls.

Penicillin fits the active site, and the enzyme starts its normal reaction. But penicillin's strained four-membered ring opens and forms a **covalent bond** with the active site, so the enzyme is **permanently** inactivated.

Without cross-links, the walls of growing bacteria are weak and the cells burst.

Key idea

Mechanism-based inhibition is irreversible: the enzyme takes part in its own inactivation.

Four ways to slow an enzyme

C1.1.14–17

Type	Binding site	Reversible?	Example
Competitive	Active site	Yes	Statins and HMG-CoA reductase
Non-competitive	Allosteric site	Yes	Many heavy metal ions
Feedback	Allosteric site	Yes	Isoleucine and threonine deaminase
Mechanism-based	Active site, covalent bond	No	Penicillin and transpeptidase

Think about it

Why must a cell replace an enzyme that has been mechanism-inhibited, but not one that was competitively inhibited?

Check-in: pathways and inhibition

C1.1.11–16

1. Give one example each of an intracellular and an extracellular enzyme reaction.
2. Explain why metabolism always releases heat.
3. Distinguish between competitive and non-competitive inhibition.
4. Explain why increasing substrate concentration reduces the effect of a statin.
5. Explain why isoleucine inhibits the first enzyme of its pathway.

How to use this

Answer without your notes, then compare with a partner.

How bacteria resist penicillin

C1.1.17

Mechanism	How it works
Changed transpeptidase	Some bacteria, such as MRSA, have a transpeptidase with a different active site. Penicillin binds poorly, but the enzyme still cross-links the wall
Penicillinase (beta-lactamase)	An enzyme that breaks penicillin's four-membered ring before it reaches the transpeptidase

Link to evolution

Bacteria with a resistance gene survive penicillin and reproduce, so the resistant strain spreads (natural selection, D4.1). Overuse of antibiotics speeds this up. Drug companies now add penicillinase inhibitors, such as clavulanic acid, to some antibiotics.

Common mistakes to avoid

Not this	But this
A non-competitive inhibitor blocks the active site.	It binds an allosteric site and changes the shape of the active site.
More substrate overcomes all inhibitors.	Only competitive inhibition is overcome by more substrate.
Feedback inhibition acts on the last enzyme.	The end product usually inhibits the first enzyme of the pathway.
Penicillin kills human cells.	It targets bacterial transpeptidases; human cells have no cell walls.
Respiration turns all the energy in glucose into ATP.	Only about a third is stored as ATP; the rest is lost as heat.

Check that you can...

1. **Compare** intracellular and extracellular enzyme reactions, with examples
2. **Explain** why metabolism releases heat, and how endotherms use it
3. **Distinguish** between linear and cyclical pathways
4. **Explain** allosteric (non-competitive) inhibition
5. **Explain** competitive inhibition, using statins
6. **Interpret** data or graphs to identify the type of inhibition
7. **Explain** feedback inhibition, using isoleucine
8. **Explain** mechanism-based inhibition by penicillin, and how resistance arises

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give one example each of an intracellular and an extracellular enzyme reaction.
2. Explain why increasing substrate concentration reduces the effect of a statin.
3. State why penicillin is called a mechanism-based inhibitor.

Image credits

Salivary alpha-amylase 1SMD.png – Own work. – Public domain

Aptenodytes forsteri -Snow Hill Island, Antarctica -adults and juvenile-8.jpg – Ian Duffy from UK – CC BY 2.0

Steam from compost heaps.jpg – Albarubescens – CC BY-SA 4.0

Penicillin core.svg – Yikrazuul – Public domain

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